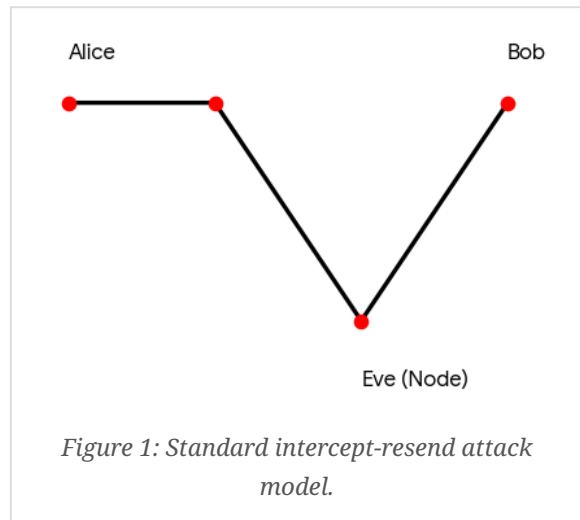


Quantum Cryptography: Comprehensive Analysis

1. Introduction

The continuous advancement of cryptographic protocols is necessitated by the rapid growth in computational power available to malicious actors. In standard public-key infrastructure, security relies entirely on mathematical complexity. However, the advent of quantum computing threatens to render these mathematical barriers obsolete via algorithms such as Shor's algorithm. To preemptively counter this vulnerability, researchers have increasingly turned to Quantum Key Distribution (QKD).



QKD leverages the fundamental laws of quantum mechanics to guarantee secure communication. By encoding information in quantum states (qubits), typically using individual photons, communicators can detect any third-party interception. This is due to the observer effect in quantum mechanics: measuring a quantum system inherently disturbs it.

2. Experimental Results

Our simulations tested the protocols over a variable fiber span. The results below highlight the critical differences in error tolerance as distance increases.

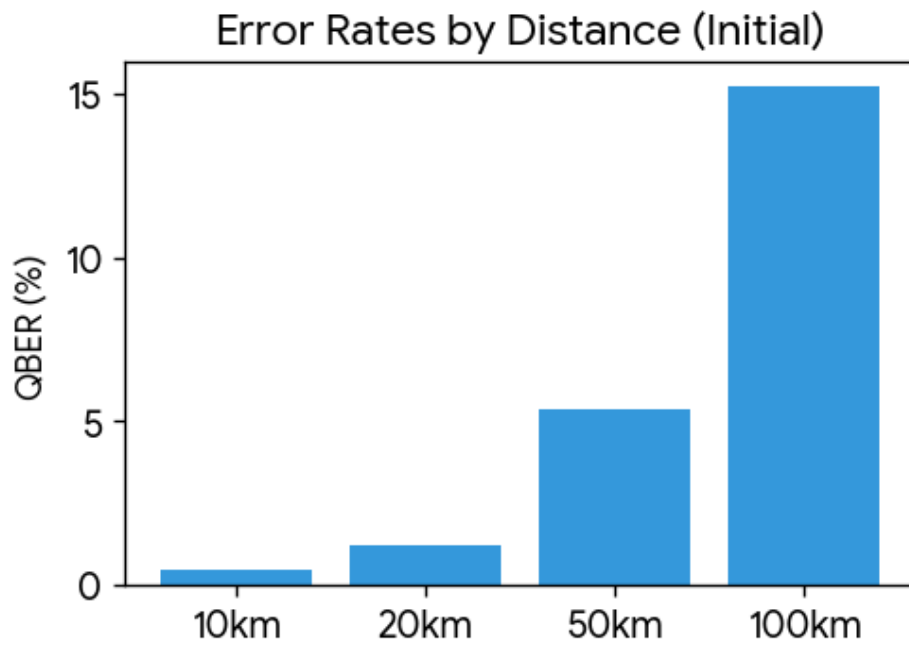


Chart 1: Initial baseline QBER vs Distance.

3. Tabular Data Summary

Distance	QBER (%)	Key Rate (kbps)
10km	0.5	500
50km	5.4	120